

Data Schema

DAC Executive Dashboard - deployed storage model and canonical analytical schema

Document	Data Schema
Application	Con Edison Disadvantaged Communities (DAC) Executive Dashboard
Source data	Con Edison annual DAC Report (2023 and 2024), filed with the NY PSC (Matter 23-02017)
Coverage	10 report sections (A-J); ~52 tables in 2024, 45 in 2023; modeled as 4 shared dimensions + 10 facts
Status	Current as of June 24, 2026 - TEST environment

1. Overview - two data models

This system involves **two distinct data models**, and it is important to keep them apart:

- **The deployed operational model (Dataverse).** What the live dashboard reads and writes today. It uses a pragmatic “grid” shape: one record per table-year, with that table’s grid stored as JSON. This is documented in section 2.
- **The canonical analytical schema (star schema).** The normalized target model that the report data is designed around - 4 shared dimensions and 10 fact tables, one fact per report section. This is the reference model for analytics and the published Schema Explorer and is documented in sections 3-5.

Known divergence: The deployed model is intentionally the grid shape (one record per table-year), not the normalized fact/dimension shape. This was a deliberate design decision for the dashboard, and reconciling the two models is an open item the team should confirm. Both are documented here so the relationship is explicit.

2. Deployed operational model (Dataverse)

The running dashboard stores everything in three Dataverse tables in the Clara Fortuna Dev environment (publisher prefix `cr2bf_`). The editable numbers do not live in `payload.json` - that file holds structure only; the numbers live here.

Display name	Logical name	Purpose
DAC Ingest Test - Table Data	<code>cr2bf_dacingesttesttabledata1</code>	One record per table-year; stores that table’s grid.
DAC Ingest Test Change History	<code>cr2bf_dacingesttestchangehistory</code>	Audit trail of every create / edit / delete.
DAC Ingest Test Reporting Year	<code>cr2bf_dacingesttestreportingyear</code>	The set of reporting years available in the UI.

2.1 Table Data - the grid record

Each record represents one table for one reporting year. The grid itself (rows and columns of numbers) is serialized as JSON in a multi-line text (memo) field, `cr2bf_rows`. This is the “grid model”: rather than one row per data point, the whole table’s grid is one record.

Conceptual field	Description
Record identity	Identifies which section/table the record belongs to (e.g. the A1, F4, J9 grid).
Reporting year	Ties the record to a year in the Reporting Year table.
<code>cr2bf_rows</code>	Memo / multiple-lines-of-text field holding the grid as a JSON array of rows. Must parse cleanly and be rectangular (consistent column count per row).

Conceptual field	Description
Standard Dataverse columns	System-managed created/modified by and on, record id, etc.

Validation rule: For every record, cr2bf_rows must be valid JSON and each grid must be rectangular. Row and column counts per table-year should match the authoritative source (payload.json / the official DAC report) before any production cutover.

2.2 Change History - the audit trail

Every create, edit, and delete writes one entry here, capturing **who** (name and email), **when** (timestamp), and **what changed**. This is the WHO/WHEN auditability record for the report.

2.3 Reporting Year

Holds the list of reporting years the dashboard exposes (e.g. 2023, 2024). Adding a year in the UI writes here, and the Table Data records reference it.

3. Canonical analytical schema (star schema)

The report data is modeled as a star schema: **4 shared dimensions** feeding **10 facts**, one fact per report section (A-J). Single-relation dimensions are “absorbed” into their fact as enum columns to keep the model compact - the dashboard’s Schema Explorer visualizes exactly this model.

Type and key conventions used in the column tables that follow:

Marker	Meaning
PK	Primary key of the table.
FK → ...	Foreign key referencing the named dimension.
ENUM	An absorbed single-relation dimension, stored as a coded enum column on the fact.

3.1 Shared dimensions

Four dimensions are reused across the facts. Sharing them is what makes cross-section analysis (e.g. “all DAC activity in Queens in 2024”) possible.

DIM_PERIOD - Time dimension

Shared across all 10 facts. period_grain distinguishes MONTH, QUARTER, YEAR, CUMULATIVE and POINT_IN_TIME. Moving from annual to monthly reporting only requires inserting rows at MONTH grain - no schema change.

Feeds: All 10 facts

Column	Type	Key / note
period_id	INTEGER	PK
period_grain	VARCHAR(16)	
period_start_date	DATE	
period_end_date	DATE	
calendar_year	INTEGER	
calendar_month	INTEGER	
calendar_quarter	INTEGER	
period_label	VARCHAR(32)	
is_report_period	BOOLEAN	

DIM_GEOGRAPHY - Hierarchical geography

Hierarchy: census tract → network / load area → borough / county → service territory. The is_dac flag tracks the NYSERDA designation at tract level.

Feeds: Outages (F), Main Replacement (G), Leak Repairs (H), Customer Operations (J)

Column	Type	Key / note
geography_id	INTEGER	PK
geography_level	VARCHAR(24)	
census_tract_geoid	VARCHAR(16)	
network_name	VARCHAR(64)	
load_area_name	VARCHAR(64)	
borough_or_county	VARCHAR(32)	
is_dac	BOOLEAN	
is_unknown_dac	BOOLEAN	

DIM_DAC_STATUS - DAC / Low-Income status

Distinguishes geographic DAC (NYSERDA census tract) from low-income (EAP participant). Sections A and J use the combined definition - which is why this is modeled as a dimension rather than a simple boolean.

Feeds: 8 facts (A, B, C, D, F, G, H, J)

Column	Type	Key / note
dac_status_id	INTEGER	PK
dac_status_code	VARCHAR(16)	
is_dac_geographic	BOOLEAN	
is_low_income_eap	BOOLEAN	
dac_status_label	VARCHAR(64)	

DIM_PROGRAM - Program catalog

All named programs with their section code, category, default segment, and flags (is_lmi_targeted, is_active).

Feeds: Sections A, B, C, I

Column	Type	Key / note
program_id	INTEGER	PK
program_code	VARCHAR(32)	
program_name	VARCHAR(128)	
section_code	VARCHAR(4)	
program_category	VARCHAR(64)	
customer_segment_default	VARCHAR(32)	
is_lmi_targeted	BOOLEAN	

Column	Type	Key / note
is_active	BOOLEAN	

4. Fact tables (one per report section)

Each fact corresponds to one section of the DAC report and produces that section's tables. Measures are the numeric/text columns; the rest are keys or absorbed enums.

FACT_CLEAN_ENERGY_SPENDING

Section A - Clean Energy Spending

Absorbed dims: customer_segment_code + measure_category_code. One row per (period × program × segment × measure category × DAC status). When measure_category_code is NULL the row feeds program-level tables (A1-A4); when populated, installation-count tables (A5-A8). Produces all of A1-A10.

Column	Type	Key / note
spending_fact_id	BIGINT	PK
period_id	INTEGER	FK → DIM_PERIOD
program_id	INTEGER	FK → DIM_PROGRAM
dac_status_id	INTEGER	FK → DIM_DAC_STATUS
customer_segment_code	VARCHAR(24)	ENUM
measure_category_code	VARCHAR(32)	ENUM
incentive_dollars	DECIMAL(18,2)	
energy_savings_mmbtu	DECIMAL(18,4)	
participants	INTEGER	
installations	INTEGER	
source_table	VARCHAR(16)	

FACT_EV_MAKE_READY

Section B - EV Make-Ready

Absorbed dim: plug_type_code (NULL for incentive-only rows in Table B1, populated for plug-count rows in B2). Produces Tables B1 and B2.

Column	Type	Key / note
ev_fact_id	BIGINT	PK
period_id	INTEGER	FK → DIM_PERIOD
program_id	INTEGER	FK → DIM_PROGRAM
dac_status_id	INTEGER	FK → DIM_DAC_STATUS
plug_type_code	VARCHAR(16)	ENUM
incentive_dollars	DECIMAL(18,2)	
plugs_installed	INTEGER	
source_table	VARCHAR(8)	

FACT_DEMAND_RESPONSE**Section C - Demand Response**

No absorbed dims. Customer-group summaries (DAC / Low-Income / Total) in Tables C2-C5 come from filtering on `dac_status_id`.

Column	Type	Key / note
<code>dr_fact_id</code>	BIGINT	PK
<code>period_id</code>	INTEGER	FK → DIM_PERIOD
<code>program_id</code>	INTEGER	FK → DIM_PROGRAM
<code>dac_status_id</code>	INTEGER	FK → DIM_DAC_STATUS
<code>participants</code>	INTEGER	
<code>committed_mw</code>	DECIMAL(10,2)	
<code>delivered_mw</code>	DECIMAL(10,2)	
<code>source_table</code>	VARCHAR(8)	

FACT_DER**Section D - Distributed Energy Resources**

Absorbed dim: `der_type_code`. `period_grain='CUMULATIVE'` handles the “Up to 2024” rollup. subscribers is NULL when `der_type_code='NM'` (Net Metering has projects, not subscribers).

Column	Type	Key / note
<code>der_fact_id</code>	BIGINT	PK
<code>period_id</code>	INTEGER	FK → DIM_PERIOD
<code>dac_status_id</code>	INTEGER	FK → DIM_DAC_STATUS
<code>der_type_code</code>	VARCHAR(8)	ENUM
<code>projects</code>	INTEGER	
<code>mw_installed</code>	DECIMAL(12,4)	
<code>subscribers</code>	INTEGER	
<code>source_table</code>	VARCHAR(8)	

FACT_STRATEGIC_CAP_INVESTMENT**Section E - Strategic Electric Capital Investments**

Absorbed dim: `investment_category_code`. `pct_affecting_dacs` stays as a row attribute (not split across DAC / non-DAC rows) because Section E reports estimated exposure, not actual dollar allocation.

Column	Type	Key / note
<code>cap_fact_id</code>	BIGINT	PK
<code>period_id</code>	INTEGER	FK → DIM_PERIOD

Column	Type	Key / note
investment_category_code	VARCHAR(24)	ENUM
total_investment_dollars	DECIMAL(18,2)	
pct_affecting_dacs	DECIMAL(5,4)	
source_table	VARCHAR(8)	

FACT_CUSTOMER_OUTAGES

Section F - Customer Outages

Absorbed dims: outage_excludability + outage_network_type (the original DIM_OUTAGE_TYPE was itself a cross-product, so absorbed as two columns). geography_id at NETWORK / LOAD_AREA feeds F4-F6; at BOROUGH feeds F8 / F9.

Column	Type	Key / note
outage_fact_id	BIGINT	PK
period_id	INTEGER	FK → DIM_PERIOD
geography_id	INTEGER	FK → DIM_GEOGRAPHY
dac_status_id	INTEGER	FK → DIM_DAC_STATUS
outage_excludability	VARCHAR(16)	ENUM
outage_network_type	VARCHAR(16)	ENUM
outage_count	INTEGER	
customers_interrupted	INTEGER	
source_table	VARCHAR(16)	

FACT_MAIN_REPLACEMENT

Section G - Main Replacement

Absorbed dim: pipe_action_code. feet_of_pipe and metric_tons_ch4_reduced coexist - emissions (G10) are derivatives of pipe replacement (G1-G9).

Column	Type	Key / note
pipe_fact_id	BIGINT	PK
period_id	INTEGER	FK → DIM_PERIOD
geography_id	INTEGER	FK → DIM_GEOGRAPHY
dac_status_id	INTEGER	FK → DIM_DAC_STATUS
pipe_action_code	VARCHAR(24)	ENUM
feet_of_pipe	INTEGER	
metric_tons_ch4_reduced	DECIMAL(10,4)	

Column	Type	Key / note
source_table	VARCHAR(8)	

FACT_LEAK_REPAIRS

Section H - Leak Repairs

Absorbed dim: leak_type_code (nullable - H1 reports an aggregate, not per-type; populating it enables per-leak-type analytics with no schema change).

Column	Type	Key / note
leak_fact_id	BIGINT	PK
period_id	INTEGER	FK → DIM_PERIOD
geography_id	INTEGER	FK → DIM_GEOGRAPHY
dac_status_id	INTEGER	FK → DIM_DAC_STATUS
leak_type_code	VARCHAR(8)	ENUM
leaks_repaired	INTEGER	
unique_tickets	INTEGER	
source_table	VARCHAR(8)	

FACT_CLEAN_ENERGY_JOBS

Section I - Clean Energy Jobs

Long-form fact. One row per (period × metric × count_basis), where count_basis ∈ {UNIQUE, NON_UNIQUE}.

Column	Type	Key / note
jobs_fact_id	BIGINT	PK
period_id	INTEGER	FK → DIM_PERIOD
program_id	INTEGER	FK → DIM_PROGRAM
metric_code	VARCHAR(32)	ENUM
count_basis	VARCHAR(16)	ENUM
count_value	INTEGER	
text_value	VARCHAR(256)	
source_table	VARCHAR(8)	

FACT_CUSTOMER_OPERATIONS

Section J - Customer Operations

Absorbed dims: account_status_code + usage_commodity_code. All J1-J9 metrics in a single long-form fact. Point-in-time metrics (J3/J4/J6/J7/J9) use period_grain='POINT_IN_TIME'; flow metrics (J5/J8) use YEAR or MONTH grain.

Column	Type	Key / note
custop_fact_id	BIGINT	PK
period_id	INTEGER	FK → DIM_PERIOD
geography_id	INTEGER	FK → DIM_GEOGRAPHY
dac_status_id	INTEGER	FK → DIM_DAC_STATUS
account_status_code	VARCHAR(32)	ENUM
usage_commodity_code	VARCHAR(16)	ENUM
account_count	INTEGER	
dollar_amount	DECIMAL(18,2)	
usage_amount	DECIMAL(18,4)	
avg_usage_per_customer	DECIMAL(12,4)	
source_table	VARCHAR(8)	

5. Source sections and year-over-year mapping

The ten report sections, and how the dashboard groups them:

Section	Title	Fact
A	Clean Energy Spending	FACT_CLEAN_ENERGY_SPENDING
B	Electric Vehicle Make-Ready	FACT_EV_MAKE_READY
C	Demand Response	FACT_DEMAND_RESPONSE
D	Distributed Energy Resources	FACT_DER
E	Strategic Electric Capital Investments	FACT_STRATEGIC_CAP_INVESTMENT
F	Customer Outages	FACT_CUSTOMER_OUTAGES
G	Main Replacement Program	FACT_MAIN_REPLACEMENT
H	Leak Repairs	FACT_LEAK_REPAIRS
I	Clean Energy Jobs	FACT_CLEAN_ENERGY_JOBS
J	Customer Operations Data	FACT_CUSTOMER_OPERATIONS

5.1 2023 ↔ 2024 mapping

The 2023 report is mapped against the 2024 canonical structure (see the [Table-by-Table Mapping](#) sheet in the source workbook). Most tables carry across as **SAME**; some are **CHANGED** (e.g. new participant types in A3/A4); and several are **NEW IN 2024** (e.g. the A7 Multisector grouping and the A9/A10 comparison summaries), which therefore have no 2023 baseline.

Status	Meaning for comparison
SAME	Table carries across both years; directly comparable (by-program may need name mapping).
CHANGED	Same table, but structure/segments changed; comparable at total level.
NEW IN 2024	Introduced in 2024; no 2023 baseline to compare against.

5.2 Cross-cutting conventions

- **DAC vs Low-Income.** “DAC” is the NYSERDA geographic (census-tract) designation; “low-income” is an EAP participant. Sections A and J use the combined definition - hence DIM_DAC_STATUS rather than a boolean.
- **Period grain.** DIM_PERIOD.period_grain carries MONTH, QUARTER, YEAR, CUMULATIVE or POINT_IN_TIME, so the same schema supports annual or monthly reporting and point-in-time vs flow metrics with no structural change.
- **Geography hierarchy.** census tract → network / load area → borough / county → service territory, with the DAC flag at tract level.

- **source_table.** Every fact carries a source_table tag recording which published table(s) a row feeds, preserving traceability back to the report.

For how these tables are read and written at runtime, see the **System Flow** document; for usage and maintenance, see the **Technical / User Documentation**.